



VP AIR TEST NEET (2025-26)

PART TEST - 08

DROPPER

DURATION: 180 Minutes

Date: 04-01-2026

M. MARKS: 720

Topics Covered

Physics:	Moving Charges and Magnetism, Magnetism and Matter
Chemistry:	Alcohols, Phenols and Ethers
Botany:	Sexual Reproduction in Flowering Plants
Zoology:	Human Reproduction

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A particle of charge q and mass m enters a region of uniform magnetic field B in vacuum at an angle $\theta = 30^\circ$ with the field lines. If the pitch of the helical path is P , what is the radius of the helix?

(1) $\frac{P}{2\pi\sqrt{3}}$ (2) $\frac{P\sqrt{3}}{2\pi}$
(3) $\frac{P}{2\pi}$ (4) $\frac{P\sqrt{3}}{\pi}$

2. Two identical short bar magnets, each of magnetic moment M , are placed such that they form a cross (+) shape with their centers coinciding. If one is along the X-axis and the other along the Y-axis (North poles at $+x$ and $+y$), the net magnetic moment of the system is:

(1) M (2) $\sqrt{2}M$
(3) $2M$ (4) Zero

3. A proton and an alpha particle enter a uniform magnetic field with the same kinetic energy, moving perpendicular to the field. What is the ratio of the radii of their circular paths ($r_p : r_\alpha$)?

(1) 1 : 1 (2) 1 : 2
(3) 2 : 1 (4) 1 : 4

4. A non-conducting ring of mass m and radius R has a charge Q distributed uniformly over it. The ring rotates about its axis with an angular velocity ω . The ratio of its magnetic moment to its angular momentum is:

(1) $\frac{Q}{m}$ (2) $\frac{Q}{2m}$
(3) $\frac{2Q}{m}$ (4) $\frac{Q}{4m}$

5. A square loop of side L carries a current I . The magnetic field at the center of the loop is B . If the same length of wire is bent into a circular loop and carries the same current, the field at the center becomes B' . The ratio $\frac{B}{B'}$ is:

(1) $\frac{8\sqrt{2}}{\pi^2}$ (2) $\frac{8}{\pi^2}$
(3) $\frac{16}{\pi^2}$ (4) $\frac{\sqrt{2}}{\pi}$

6. A paramagnetic material has a susceptibility $\chi = 0.001$ at $300K$. What will be its susceptibility at $600K$?

(1) 0.0005 (2) 0.002
(3) 0.001 (4) 0.00025

7. A long straight wire of radius R carries a current I distributed non-uniformly such that the current density $J = kr^2$, where r is the distance from the axis. The magnetic field at $r = R/2$ is:

(1) $\frac{\mu_0 k R^3}{16}$ (2) $\frac{\mu_0 k R^3}{32}$
(3) $\frac{\mu_0 k R^3}{8}$ (4) $\frac{\mu_0 k R^2}{4}$

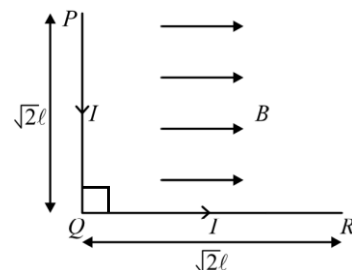
8. An electron moves in a circle of radius 1.0cm with a speed of $6 \times 10^6 \text{m/s}$. The magnetic field at the center of the circle is approximately:

(1) 10^{-4}T
(2) $9.6 \times 10^{-16} \text{T}$
(3) $1.2 \times 10^{-5} \text{T}$
(4) $4.8 \times 10^{-16} \text{T}$

9. A sample of magnetic material is placed in a magnetizing field of $2 \times 10^3 \text{A/m}$. If the magnetic susceptibility of the material is 599, what is the magnetic induction (B) inside the material? (Take $\mu_0 = 4\pi \times 10^{-7} \text{T} \cdot \text{m/A}$)

(1) $0.24\pi \text{T}$ (2) $0.48\pi \text{T}$
(3) $1.2\pi \text{T}$ (4) $2.4\pi \text{T}$

10. A wire PQR is placed in a region of uniform magnetic field B as shown in the figure. A current I flows through the wire. The magnitude of force on wire PQR will be: (wire PQR and magnetic field are in same plane)



(1) $2IlB$ (2) IlB
(3) $\sqrt{2}IlB$ (4) $4IlB$

11. Two infinitely long parallel wires are separated by distance d and carry currents I and $2I$ in the same direction. The location of the point between them where the magnetic field is zero is:
- $\frac{d}{3}$ from wire I
 - $\frac{d}{3}$ from wire $2I$
 - $\frac{d}{4}$ from wire I
 - $\frac{d}{2}$ from wire I
12. A material has relative permeability $\mu_r = 0.9999$. This material is:
- Diamagnetic
 - Paramagnetic
 - Ferromagnetic
 - Non-magnetic
13. In a moving coil galvanometer, the deflection ϕ is related to the current I as:
- $I \propto \phi$
 - $I \propto \tan\phi$
 - $I \propto \sin\phi$
 - $I \propto \phi^2$
14. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): The magnetic field lines form closed loops.
Reason (R): Isolated magnetic monopoles do not exist.
 In the light of the above statements, choose the **correct** answer from the options given below:
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true and R is NOT the correct explanation of A.
15. A current of $10A$ flows through a semi-circular wire of radius $2cm$. The magnetic field at the center of the arc is:
- $1.57 \times 10^{-4} T$
 - $3.14 \times 10^{-4} T$
 - $10^{-4} T$
 - $6.28 \times 10^{-4} T$
16. The magnetic field at a point P on the axis of a circular coil is $\frac{1}{8}$ of the value at the center. If the radius of the coil is R , the distance of P from the center is:
- $R\sqrt{3}$
 - $R\sqrt{2}$
 - $3R$
 - $2R$
17. A domain in a ferromagnetic substance is in the form of a cube of side $1\mu m$. If it contains 10^{10} atoms and each atomic dipole moment is $1.5\mu_B$ (where $\mu_B = 9.27 \times 10^{-24} Am^2$), the magnetization M of the domain is:
- $1.4 \times 10^5 A/m$
 - $1.4 \times 10^9 A/m$
 - $1.4 \times 10^{11} A/m$
 - $9.27 \times 10^4 A/m$
18. Two particles X and Y having equal charges are accelerated through the same potential difference and then enter a region of uniform magnetic field, describing circular paths of radii R_1 and R_2 . The ratio of their masses $\frac{m_1}{m_2}$ is:
- $\left(\frac{R_1}{R_2}\right)^2$
 - $\frac{R_1}{R_2}$
 - $\left(\frac{R_2}{R_1}\right)^2$
 - $\sqrt{\frac{R_1}{R_2}}$
19. A wire of length L carries current I along the positive X -axis. A magnetic field $\vec{B} = B_0(\hat{i} + \hat{j} + \hat{k})$ exists. The magnitude of the magnetic force acting on the wire is:
- $\sqrt{3}ILB_0$
 - $\sqrt{2}ILB_0$
 - ILB_0
 - Zero
20. A long solenoid of length $0.5m$ and radius $1cm$ has 500 turns. It carries a current of $5A$. The magnetic field at the end of the solenoid is:
- $3.14 \times 10^{-3} T$
 - $6.28 \times 10^{-3} T$
 - $1.57 \times 10^{-3} T$
 - $9.42 \times 10^{-3} T$

21. Consider a bar magnet placed in region of uniform magnetic field such that magnetic moment of bar magnet is along the direction of magnetic field. If net work done in rotating it by an angle of 90° is 'n' times the work done in rotating it by angle of 60° then 'n' is equal to:

- (1) $\frac{1}{2}$ (2) 2
(3) $\frac{1}{4}$ (4) 1

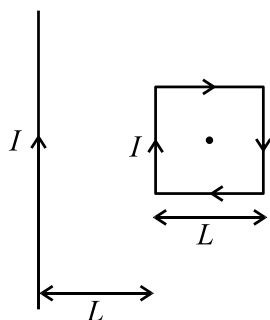
22. A rectangular loop of dimensions $10\text{ cm} \times 5\text{ cm}$ carries a current of 2A. It is placed in a magnetic field of 0.1 T such that the normal to the loop makes 30° with the field. The torque is:

- (1) $5 \times 10^{-4}\text{ N-m}$
(2) 10^{-3} N-m
(3) $8.66 \times 10^{-4}\text{ N-m}$
(4) $2.5 \times 10^{-4}\text{ N-m}$

23. A square frame of side a carries a current I . It is placed in a uniform magnetic field B such that the plane of the frame is parallel to the field. The torque is:

- (1) Ia^2B (2) $2Ia^2B$
(3) Zero (4) $Ia^2B/2$

24. A long straight wire carries a current I . A square loop of side L also carrying a steady current I is placed with two sides parallel to the wire as shown in figure. The distance of the nearest side from the wire is L . The magnitude of net force on the loop is:



- (1) $\frac{\mu_0 I^2}{2\pi}$ (2) $\frac{\mu_0 I^2}{4\pi}$
(3) $\frac{\mu_0 I^2}{6\pi}$ (4) $\frac{\mu_0 I^2}{8\pi}$

25. Consider a charged particle moving in a uniform magnetic field.

- (i) Its kinetic energy remains constant.
(ii) Its acceleration is always constant and non-zero.
(iii) Its path can be a helix.
(iv) The magnetic force does no work.

Which are **correct**?

- (1) (i), (iii), (iv)
(2) (i), (ii), (iii)
(3) (ii), (iii), (iv)
(4) All are correct.

26. The magnetic field at a perpendicular distance r from a long straight wire is $0.4G$. The field at distance $2r$ is: ($G \rightarrow$ gauss)

- (1) $0.1G$
(2) $0.2G$
(3) $0.8G$
(4) $1.6G$

27. Two identical coils carry the same current I . They are placed co-axially with centers at $x=0$ and $x=d$. The field at the midpoint is B . If the current in one is reversed, the field at the midpoint becomes:

- (1) B (2) $2B$
(3) Zero (4) $\frac{B}{2}$

28. An electron is moving with speed v in a circle of radius r . It has a magnetic moment M . If the speed is doubled and the radius is halved, the new magnetic moment will be:

- (1) M
(2) $2M$
(3) $4M$
(4) $\frac{M}{2}$

29. Let $\vec{B}_1$ be the magnetic field at a point on the axial line and $\vec{B}_2$ be the magnetic field at a point on the equatorial line of a bar magnet (both points are equidistant from the center). What is the angle between the vectors $\vec{B}_1$ and $\vec{B}_2$?

- (1) 0° (2) 90°
(3) 180° (4) 45°

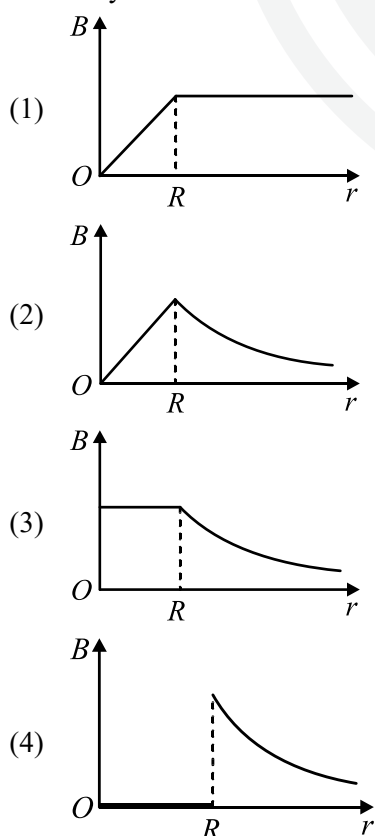
30. Inside a magnetic substance, the total magnetic induction ($\vec{B}$) is the result of both the external magnetizing field ($\vec{H}$) and the material's internal response (magnetization, $\vec{M}$). If the magnetic susceptibility of the material is denoted by χ , which of the following correctly expresses the total magnetic induction ($\vec{B}$) in terms of the applied magnetizing field ($\vec{H}$)?

(1) $B = \mu_0 H \chi$ (2) $B = \mu_0 H (1 + \chi)$
 (3) $B = \mu_0 \frac{H}{\chi}$ (4) $B = \mu_0 H (1 - \chi)$

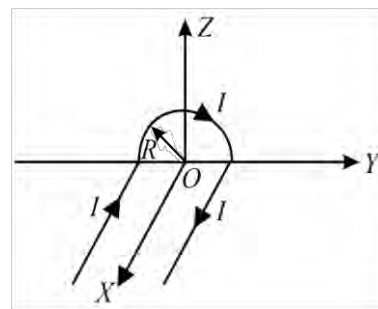
31. In a certain region, a uniform electric field $\vec{E} = 20\hat{j}$ V/m and a uniform magnetic field $\vec{B} = 5\hat{k}$ T coexist. A charge q is projected with velocity $\vec{v}$ such that it moves undeflected. The velocity $\vec{v}$ is:

(1) $4\hat{i}$ m/s (2) $-4\hat{i}$ m/s
 (3) $100\hat{i}$ m/s (4) $-100\hat{i}$ m/s

32. An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface. Choose the **correct** representation of magnetic field B as a function of radial distance r from the axis of cylinder.



33. A wire carrying current I has the shape as shown in adjoining figure.



Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying in Y-Z plane. Magnetic field at point O is;

(1) $\vec{B} = -\frac{\mu_0 I}{4\pi R} (\pi\hat{i} + 2\hat{k})$
 (2) $\vec{B} = \frac{\mu_0 I}{4\pi R} (\pi\hat{i} - 2\hat{k})$
 (3) $\vec{B} = \frac{\mu_0 I}{4\pi R} (\pi\hat{i} + 2\hat{k})$
 (4) $\vec{B} = -\frac{\mu_0 I}{4\pi R} (\pi\hat{i} - 2\hat{k})$

34. Curie temperature is the specific temperature above which:

- (1) A paramagnetic material becomes diamagnetic.
 (2) A ferromagnetic material becomes paramagnetic.
 (3) A diamagnetic material becomes paramagnetic.
 (4) A ferromagnetic material becomes diamagnetic.

35. A bar magnet of length 0.10 m is having magnetic moment 12 Am^2 . Magnetic field at a point on its axis a distance 20 cm from its centre is:

(1) $3.4 \times 10^{-4} \text{ T}$
 (2) $8 \times 10^{-5} \text{ T}$
 (3) $2 \times 10^{-5} \text{ T}$
 (4) $4 \times 10^{-5} \text{ T}$

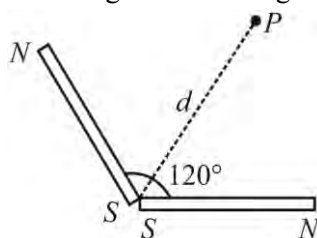
36. Torque acting on a bar magnet of length 20 cm, magnetic dipole moment $4 \times 10^{-6} \text{ Am}^2$, which is placed in uniform magnetic field $2 \times 10^{-5} \text{ T}$ perpendicular to the field is:

(1) $3 \times 10^{-11} \text{ Nm}$
 (2) $2 \times 10^{-11} \text{ Nm}$
 (3) $8 \times 10^{-11} \text{ Nm}$
 (4) 0 Nm

37. A particle of specific charge (charge-to-mass ratio) α is projected from the origin at $t=0$ with an initial velocity $\vec{v}_0 = v_0(\hat{i} + \hat{j})$ in a region of space where a uniform magnetic field $\vec{B} = B_0\hat{j}$ exists. Neglecting all other forces, the velocity vector of the particle at the instant $t = \frac{\pi}{\alpha B_0}$ will be:

- (1) $v_0(\hat{i} + \hat{j})$ (2) $v_0(-\hat{i} + \hat{j})$
 (3) $v_0(\hat{j} + \hat{k})$ (4) $v_0(\hat{j} - \hat{k})$

38. Two identical short bar magnets are placed at 120° as shown in the figure. The magnetic moment of each magnet is M . Then the magnetic field at the point P on the angle bisector is given by:



- (1) $\frac{\mu_0}{4\pi} \cdot \frac{M}{d^3}$ (2) $\frac{\mu_0}{4\pi} \cdot \frac{2M}{d^3}$
 (3) $\frac{\mu_0}{4\pi} \cdot \frac{2\sqrt{2}M}{d^3}$ (4) zero

39. A short bar magnet of magnetic moment 0.32 J/T is placed in uniform field of 0.15 T . What is the potential energy of bar magnet in stable equilibrium condition?

- (1) $-3.2 \times 10^{-2} \text{ J}$ (2) $-2.5 \times 10^{-2} \text{ J}$
 (3) $-4.8 \times 10^{-2} \text{ J}$ (4) $-1.5 \times 10^{-2} \text{ J}$

40. A magnetic force acting on a charged particle is given by $(-8\hat{k})\text{N}$. The velocity of the particle is $(2\hat{i} + 3\hat{j}) \times 10^6 \text{ m/s}$ and the magnetic field of 2 T is applied in the $+y$ direction. Find the charge on the particle.

- (1) $2\mu\text{C}$ (2) $3\mu\text{C}$
 (3) $-2\mu\text{C}$ (4) $-3\mu\text{C}$

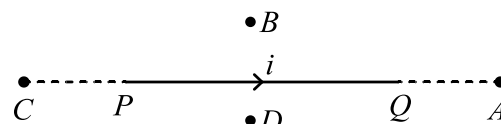
41. A sample of material is placed in an external magnetic field. It is observed that the material is feebly repelled by the field and moves from the stronger to the weaker part of the field. This material is:

- (1) Paramagnetic (2) Ferromagnetic
 (3) Diamagnetic (4) Non-magnetic

42. A small circular flexible loop of wire of radius R carries a current I . It is placed in a uniform perpendicular magnetic field B . The tension in the loop will be doubled, if;

- (1) I is doubled and B, R are constant
 (2) B is doubled and I, R is kept constant
 (3) R is doubled and I, B are constant
 (4) All of these

43. A current i is flowing in a straight wire PQ as shown in figure.



Magnetic field due to current carrying wire PQ is zero at point:

- (1) A and D (2) B and D
 (3) A and C (4) B and C

44. A small N -turn coil of area A carrying current i is placed inside a long solenoid (n turns/m, current I) such that their axes are perpendicular. The torque on the coil is:

- (1) $\mu_0 n I N i A$ (2) $\frac{1}{2} \mu_0 n I N i A$
 (3) Zero (4) $2\mu_0 n I N i A$

45. Match List-I with List-II.

List-I (Magnetic material)		List-II (Value of relative magnetic permeability μ_r)	
(A)	Diamagnetic	(I)	$\mu_r \gg 1$ (Very large)
(B)	Paramagnetic	(II)	$0 \leq \mu_r < 1$ (Less than unity)
(C)	Ferromagnetic	(III)	$\mu_r > 1$ (Slightly greater than unity)

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III
 (2) A-II, B-III, C-I
 (3) A-III, B-II, C-I
 (4) A-II, B-I, C-III

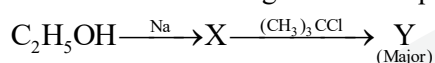
46. The compound which react fastest with Lucas reagent at room temperature is:

- (1) Propan-1-ol
- (2) 2-Methylpropan-2-ol
- (3) Propan-2-ol
- (4) Ethanol

47. The reaction occurring by breaking of C–O bond in propanol is:

- (1) Reaction with SOCl_2
- (2) Reaction with Na
- (3) Reaction with CH_3COOH
- (4) Reaction with KMnO_4

48. Consider the following reaction sequence:



The product (Y) is:

- (1) $\text{C}_2\text{H}_5-\text{O}-\text{C}(\text{CH}_3)_3$
- (2) $\text{C}_2\text{H}_5-\text{C}(\text{CH}_3)_3$
- (3) $(\text{CH}_3)_2\text{C}=\text{CH}_2$
- (4) $(\text{CH}_3)_3\text{C}-\text{CH}_2\text{Cl}$

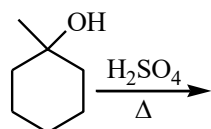
49. Match List-I with List-II.

List-I (Compound)		List-II (pK _a value)	
(A)	m-Nitrophenol	(I)	15.9
(B)	p-Cresol	(II)	10
(C)	Phenol	(III)	10.2
(D)	Ethanol	(IV)	8.3

Choose the **correct** answer from the options given below:

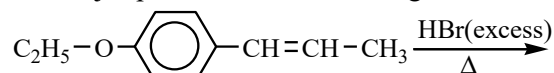
- (1) A-III; B-IV; C-II; D-I
- (2) A-II; B-IV; C-I; D-III
- (3) A-I; B-II; C-IV; D-III
- (4) A-IV; B-III; C-II; D-I

50. Major product of the following reaction is:



- (1)
- (2)
- (3)
- (4)

51. The major product of the following reaction is:



- (1)
- (2)
- (3)
- (4)

52. Today, most of the methanol is produced by catalytic hydrogenation of:

- (1) CH_2O
- (2) CO
- (3) C_3O_2
- (4) CO_2

53. Major product expected from the mononitration of phenylethanoate is:

- (1)
- (2)
- (3)
- (4)

54. **Correct** statements regarding Reimer-Tiemann reaction are:

- A. It is an example of electrophilic substitution reaction
 - B. Salicylaldehyde is obtain as the major product
 - C. The intermediate is substituted benzal chloride
- (1) A and B only
 - (2) B and C only
 - (3) A and C only
 - (4) A, B and C

55. Phenol on reduction with Zn dust gives:

- (1) Alicyclic compound
- (2) Aromatic compound
- (3) Non-benzenoid compound
- (4) Non-aromatic compound

56. IUPAC name of methyl isopropyl ether is:

- (1) 1-Methoxypropane
- (2) 2-Ethoxypropane
- (3) 2-Methoxypropane
- (4) 1-Ethoxypropane

57. Match List-I with List-II.

List-I		List-II	
(A)	Anisole	(I)	
(B)	Phenetole	(II)	
(C)	Catechol	(III)	
(D)	Quinol	(IV)	

Choose the **correct** answer from the options given below:

- (1) A-III; B-IV; C-II; D-I
- (2) A-II; B-IV; C-I; D-III
- (3) A-IV; B-I; C-II; D-III
- (4) A-III; B-I; C-IV; D-II

58. Which of the following compounds give picric acid on treatment with concentrated nitric acid?

- (1)
- (2)
- (3)
- (4)

59. Anisole on reaction with ethanoyl chloride in presence of anhyd. AlCl_3 mainly gives majorly:

- (1)
- (2)
- (3)
- (4)

60. Most reactive hydrogen halide with ethers is:

- (1) HF
- (2) HBr
- (3) HCl
- (4) HI

61. The major product of the following reaction is:



- (1) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2\text{OH}$
- (2) $\text{CH}_3-\text{CH}_2\text{CH}_2\text{COOH}$
- (3) $\text{CH}_3-\text{CH}=\text{CH}-\text{COOH}$
- (4) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2\text{OH}$

62. Glucose and fructose undergo fermentation to give ethanol in presence of enzyme:

- (1) Invertase
- (2) Zymase
- (3) Diastase
- (4) Maltase

63. Given below are two statements:

Statement I: A better reagent for oxidation of primary alcohols to aldehydes in good yield is PCC.

Statement II: Secondary alcohols are oxidised to ketones by CrO_3

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

64. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Picric acid give CO_2 on reaction with NaHCO_3 .

Reason R: Picric acid is more acidic than carbonic acid.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

65. Given below are two statements:

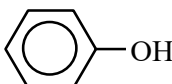
Statement I: Most of the worldwide production of phenol is from cumene.

Statement II: Diazonium salts are hydrolysed to benzene by warming with water.

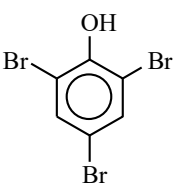
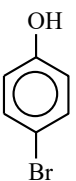
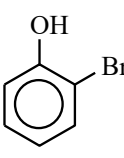
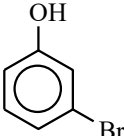
In the light of the above statements, choose the *most appropriate* answer from the options given below:

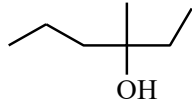
- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

66. Least acidic compound among the following is:

- (1) CH_3COOH
- (2) H_2O
- (3) 
- (4) $\text{CH}_3\text{CH}_2\text{OH}$

67. The major product obtain when phenol is treated with Br_2 in CS_2 at 273 K is:

- (1) 
- (2) 
- (3) 
- (4) 

68.  can be prepared by the reaction of:

- A. $\text{C}_2\text{H}_5\text{COCH}_3 + \text{CH}_3\text{CH}_2\text{CH}_2\text{MgCl}$ followed by hydrolysis
 - B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COCH}_3 + \text{C}_2\text{H}_5\text{MgBr}$ followed by hydrolysis
 - C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COC}_2\text{H}_5 + \text{CH}_3\text{MgCl}$ followed by hydrolysis
- (1) A and B only
 - (2) B and C only
 - (3) A and C only
 - (4) A, B and C

69. Major product of the following reaction is:



- (1) $\text{CH}_3-\text{CH}=\text{CH}_2$
- (2) $\text{CH}_3-\text{CH}_2-\text{CHO}$
- (3) $\text{CH}_3-\text{CH}_2-\text{COOH}$
- (4) $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$

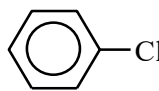
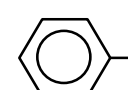
70. Phenol on oxidation with chromic acid produces:

- (1) Benzoquinone
- (2) Hydraquinone
- (3) Benzophenone
- (4) Acetophenone

71. The main reaction product of Kolbe's reaction is:

- (1) 2-Hydroxybenzoic acid
- (2) Acetylsalicylic acid
- (3) Phenyl salicylate
- (4) Methyl salicylate

72. Which of the following is most appropriate reaction for the preparation of ether?

- (1) $(\text{CH}_3)_3\text{CONa} + \text{CH}_3\text{Cl}$
- (2) $\text{CH}_3\text{ONa} +$ 
- (3)  + $(\text{CH}_3)_2\text{CHCl}$
- (4) $\text{C}_2\text{H}_5\text{ONa} + (\text{CH}_3)_3\text{CCl}$

73. Ethanol is presence of sulphuric acid at 413 K mainly gives:

- (1) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
- (2) $\text{CH}_2=\text{CH}_2$
- (3) $(\text{C}_2\text{H}_5)_2\text{SO}_4$
- (4) $\text{C}_2\text{H}_5\text{HSO}_4$

74. Given below are two statements:

Statement I: Tertiary alcohols do not undergo oxidation reactions.

Statement II: The commercial alcohol is made unfit for drinking by mixing in it aniline and zinc sulphate.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

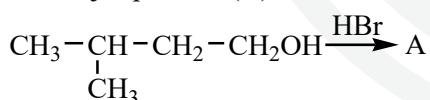
75. Given below are two statements:

Statement I: Phenol is more reactive than phenoxide ion towards electrophilic aromatic substitution.

Statement II: Ethanol is also known as wood spirit. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

76. The major product (A) of the following reaction is:



- (1) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$
- (2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_2\text{Br}$
- (3) $\text{CH}_3-\underset{\text{CH}_2\text{Br}}{\text{CH}}-\text{CH}_2-\text{CH}_3$
- (4) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{Br}}{\text{C}}}-\text{CH}_2-\text{CH}_3$

77. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The C–O–C, bond angle in CH_3OCH_3 is slightly less than the tetrahedral angle.

Reason R: There is repulsive interaction between the two lone pair of oxygen atoms.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

78. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: 2-Methylpropan-2-ol undergo dehydration more easily than butan-1-ol with concentrated H_2SO_4 at 170°C .

Reason R: 3° -Carbocation is more stable than 1° or 2° -carbocation.

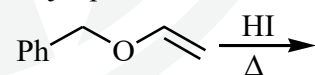
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

79. Most suitable method for the separation of o-nitrophenol and p-nitrophenol is:

- (1) Vacuum distillation
- (2) Steam distillation
- (3) Fractional distillation
- (4) Chromatographic separation

80. Major products of the following reaction is:



- (1) $\text{PhCH}_2\text{OH} + \text{CH}_2=\text{CH}-\text{I}$
- (2) $\text{PhCH}_2\text{I} + \text{CH}_2=\text{CH}-\text{I}$
- (3) $\text{PhCH}_2\text{I} + \text{CH}_3\text{CHO}$
- (4) $\text{PhCH}_2\text{I} + \text{CH}_3\text{CH}_2\text{OH}$

81. **Incorrect** statement regarding methanol is:

- (1) ingestion of it can cause blindness
- (2) it is used for making formaldehyde
- (3) it was produced by destructive distillation of wood.
- (4) it is also known as carboic acid

82. **Correct** order of acidic nature of alcohols is:

- (1) $1^\circ > 2^\circ > 3^\circ$
- (2) $3^\circ > 2^\circ > 1^\circ$
- (3) $3^\circ > 1^\circ > 2^\circ$
- (4) $2^\circ > 3^\circ > 1^\circ$

83. Given below are two statements:

Statement I: The solubility of alcohols in water decreases with increase in size of alkyl group.

Statement II: Alcohols react both as nucleophiles and electrophile.

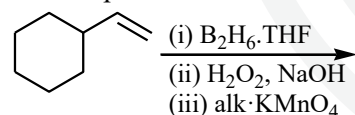
In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

84. In the cumene to phenol preparation in the presence of air, the intermediate is:

- (1)
- (2)
- (3)
- (4)

85. The final product obtained in the following multistep reaction is:

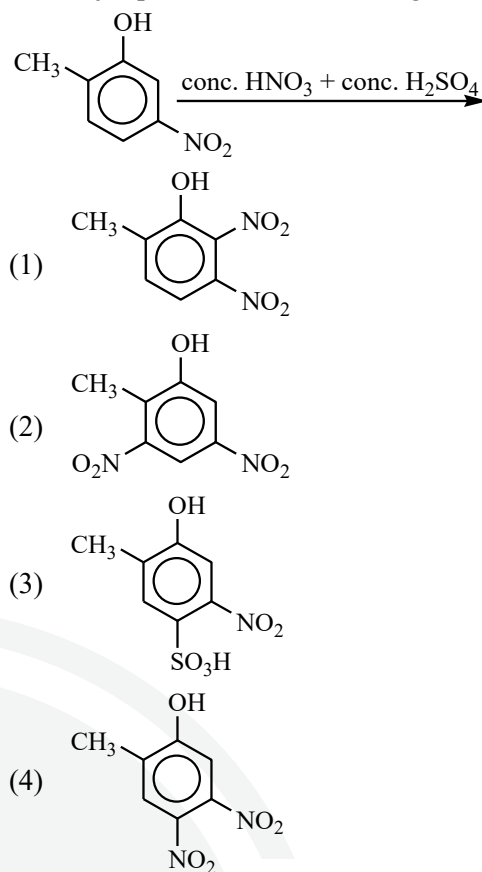


- (1)
- (2)
- (3)
- (4)

86. The gas evolve on treating CH_3MgBr with ethanol is:

- (1) Methane
- (2) Ethane
- (3) Propane
- (4) Methanal

87. The major product of the following reaction is:



88. The reaction of phenol with benzoyl chloride to give phenylbenzoate is known as:

- (1) Schotten-Baumann reaction
- (2) Kolbes reaction
- (3) Etard reaction
- (4) Gattermann Koch reaction

89. **Correct** statements regarding Williamsons synthesis are:

- (A) Follow $\text{S}_{\text{N}}2$ pathway
 - (B) Unsymmetrical ethers can be prepared
 - (C) Phenyl alkyl ethers can not be prepared
- (1) A and B only (2) B and C only
(3) A and C only (4) A, B and C

90. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Boiling point of ethanol is more than that of boiling point of methoxy ethane.

Reason R: It is due to presence of the intramolecular hydrogen bonding in ethanol.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

91. Which of the following is a vital process that enables the creation of new variants, thereby enhancing the survival advantage of a species?

- (1) Asexual reproduction
- (2) Sexual reproduction
- (3) Vegetative reproduction
- (4) Fragmentation

92. Given below are two statements:

Statement I: A typical angiosperm anther is bilobed, with each lobe having two theca.

Statement II: The anther is a tetragonal structure consisting of two microsporangia, one in each lobe. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

93. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Perisperm	(I)	Wall of fruit
(B)	Pericarp	(II)	Remnant of second cotyledon in grass embryo
(C)	Epiblast	(III)	Residual, persistent nucellus in seed
(D)	Epicotyl	(IV)	Portion of embryonal axis above the level of cotyledons in dicot embryo

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-II, C-III, D-I
- (3) A-I, B-III, C-II, D-IV
- (4) A-III, B-IV, C-I, D-II

94. Identify the **correct** statements about Panchanan Maheshwari from the following.

- A. He established the Department of Botany, University of Delhi.
- B. He popularised the use of embryological characters in taxonomy.
- C. He emphasised the need for initiation of work on artificial culture of mature embryos.

D. He worked on test tube fertilisation and intra-ovarian pollination.

E. He was honoured with a fellowship of the Royal Society of London (FRS).

Choose the **correct** option.

- (1) A, B and C only
- (2) A, B, D and E only
- (3) C, D and E only
- (4) A, B, C, D and E

95. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Cleistogamous flowers invariably produce an assured seed-set even in the absence of pollinators.

Reason (R): Cleistogamous flowers do not open at all, and the anthers and stigma lie close to each other. In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

96. In a typical dicot embryo, the cylindrical portion below the level of cotyledons that terminates in the root tip is called:

- (1) coleoptile.
- (2) plumule.
- (3) hypocotyl.
- (4) coleorhiza.

97. In the context of artificial hybridisation in a plant with bisexual flowers, the **correct** sequence of events/steps is:

- (1) Bagging → Emasculation → Pollination → Rebagging
- (2) Emasculation → Bagging → Pollination → Rebagging
- (3) Pollination → Emasculation → Bagging → Rebagging
- (4) Emasculation → Pollination → Bagging → Rebagging

98. Identify the **correctly** matched pair.

- (1) Non-albuminous seed - Castor
- (2) False fruit - Mango
- (3) Parthenocarpic fruit - Banana
- (4) Albuminous seed - Pea

99. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In angiosperms, endosperm development precedes embryo development.

Reason (R): The developing embryo in an angiosperm obtains nutrition from the endosperm.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

100. In over 60 percent of angiosperms, pollen grains are shed at the:

- (1) 2-celled stage.
- (2) 3-celled stage.
- (3) 4-celled stage.
- (4) 1-celled stage.

101. The ability of the pistil to recognise the compatible or incompatible pollen is a result of:

- (1) a physical interaction between pollen and pistil.
- (2) a continuous dialogue mediated by chemical components.
- (3) the secretion of hormones by the synergids.
- (4) the filiform apparatus guiding any pollen tube.

102. Consider the following statements about pollination.

- A. Pollination by water is quite common in flowering plants.
- B. Wind-pollinated flowers often have a single ovule in each ovary.
- C. In *Vallisneria*, the female flower reaches the surface of the water by a long stalk.
- D. Both wind and water pollinated flowers are very colourful and produce nectar.
- E. In water hyacinth and water lily, flowers are pollinated by insects or wind.

Choose the option with the **correct** statements.

- (1) A, B and C only
- (2) B, C and E only
- (3) A, D and E only
- (4) C, D and E only

103. In a flowering plant, if the number of chromosomes in a leaf cell is 20, what would be the number of chromosomes in the primary endosperm nucleus (PEN)?

- (1) 10
- (2) 20
- (3) 30
- (4) 40

104. Given below are two statements:

Statement I: Geitonogamy is functionally cross-pollination involving a pollinating agent.

Statement II: Genetically, geitonogamy is similar to autogamy since the pollen grains come from the same plant.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

105. The phenomenon where an embryo develops from a diploid egg cell formed without reduction division is a type of:

- (1) parthenocarpy.
- (2) polyembryony.
- (3) apomixis.
- (4) parthenogenesis.

106. The hard outer layer of the pollen grain, called exine, is made up of:

- (1) pectin.
- (2) cellulose.
- (3) sporopollenin.
- (4) lignin.

107. What is the function of the filiform apparatus present in the synergids?

- (1) It nourishes the egg cell.
- (2) It produces nectar to attract pollinators.
- (3) It guides the entry of the pollen tube into synergids.
- (4) It forms the wall of the pollen tube.

- 108.** Identify the **correct** statements outbreeding devices.
- They are mechanisms to discourage self-pollination.
 - Pollen release and stigma receptivity is non-synchronised is one such device.
 - Monoecious condition prevents geitonogamy.
 - Self-incompatibility is a genetic mechanism to prevent fertilisation by self-pollen.
 - Placement of anther and stigma at different positions is a device to prevent autogamy.

Choose the **correct** option.

- A, B, C, D and E
- A, B and E only
- A, B, D and E only
- C, D and E only

- 109.** Match **List-I** with **List-II**.

List-I (Structure)		List-II (Function/Feature)	
(A)	Tapetum	(I)	Protective layer of microsporangium wall, helps in anther dehiscence
(B)	Sporogenous tissue	(II)	Possesses potential pollen mother cells
(C)	Endothecium	(III)	Layer of microsporangium wall that nourishes developing pollen grains
(D)	Germ pore	(IV)	Aperture in pollen grain where sporopollenin is absent

Choose the *most appropriate* answer from the options given below.

- A-IV, B-III, C-II, D-I
 - A-IV, B-II, C-III, D-I
 - A-III, B-II, C-I, D-IV
 - A-III, B-IV, C-I, D-II
- 110.** The number of male gametes that would eventually form from one pollen mother cell is;
- two only.
 - eight only.
 - ten.
 - six only.
- 111.** Select the **incorrectly** matched pair.
- Pollination by water - *Zostera*
 - Biotic pollinator - Lemur
 - Wind pollination - Grasses
 - Pollination by insects - *Hydrilla*

- 112.** Which of the following sets has both parasitic plants?

- Orobanche* and *Striga*
- Paddy and mango
- Citrus* and wheat
- Orange and watermelon

- 113.** The ears of a corn cob;

- are adapted for water-pollination.
- are parts of female reproductive organ.
- are surrounded by a mucilaginous covering.
- have no role in sexual reproduction.

- 114.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

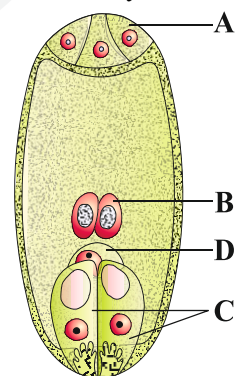
Assertion (A): It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-196°C).

Reason (R): Pollen grains are well-preserved as fossils.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

- 115.** Identify the labels A, B, C, and D in the given diagram of a mature embryo sac.



- A-Egg, B-Antipodals, C-Synergids, D-Polar nuclei
- A-Synergids, B-Polar nuclei, C-Egg, D-Antipodals
- A-Antipodals, B-Egg, C-Polar nuclei, D-Synergids
- A-Antipodals, B-Polar nuclei, C- Synergids, D-Egg

- 116.** In some cereals like rice and wheat, the pollen grains lose viability:
- (1) within 30 minutes of their release.
 - (2) after several months.
 - (3) in about 60 minutes.
 - (4) after 24 hours.
- 117.** Identify the **correct** statements regarding the structure and development of female gametophyte in angiosperms.
- A. In a majority of flowering plants, monosporic development takes place.
 - B. The megaspore mother cell undergoes mitotic divisions to produce four megaspores.
 - C. In a majority of flowering plants, three of the four megaspores degenerate.
 - D. The nucleus of the functional megaspore divides mitotically to form eight nuclei.
 - E. A typical mature embryo sac is 8-celled and 7-nucleate.
- (1) A, C and D only
 - (2) B, C and E only
 - (3) A, D and E only
 - (4) A, B and C only
- 118.** What do the terms syngamy and triple fusion refer to, respectively?
- (1) Fusion of one male gamete with the nucleus of egg cell; fusion of another male gamete with synergids.
 - (2) Fusion of one male gamete with the nucleus of egg cell; fusion of another male gamete with the two polar nuclei.
 - (3) Fusion of two male gametes with the nucleus of egg cell; fusion of one male gamete with the two polar nuclei.
 - (4) Fusion of two male gametes with polar nuclei; fusion of another male gamete with the nucleus of egg cell.
- 119.** The basal bulged part of the pistil is the:
- (1) stigma.
 - (2) style.
 - (3) ovary.
 - (4) thalamus.
- 120.** Identify the **correct** statement.
- (1) A monocarpellary syncarpous gynoecium is present in *Papaver*.
 - (2) The body of the ovule fuses with funicle in the region called micropyle.
 - (3) The cells of nucellus are haploid.
 - (4) A multicarpellary, apocarpous gynoecium is present in *Michelia*.

- 121.** Given below are two statements:

Statement I: If seeds are made into apomicts, the hybrid progeny will not show segregation of characters.

Statement II: Apomictic seeds have to be produced every year.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

- 122.** Which of the following does **not** produce both the closed and open flowers?

- (1) Common pansy
- (2) *Commelina*
- (3) Papaya
- (4) *Oxalis*

- 123.** Match **List I** with **List II**.

List-I		List-II	
(A)	<i>Vallisneria</i>	(I)	Water-pollinated marine plant
(B)	Seagrasses	(II)	Flower is 6 feet in height
(C)	<i>Yucca</i>	(III)	Water-pollinated fresh-water plant
(D)	<i>Amorphophallus</i>	(IV)	Gets pollinated by moth

Choose the *most appropriate* answer from the options given below.

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-IV, D-III
- (4) A-III, B-I, C-IV, D-II

- 124.** How do flowers pollinated by flies and beetles usually attract these animals?

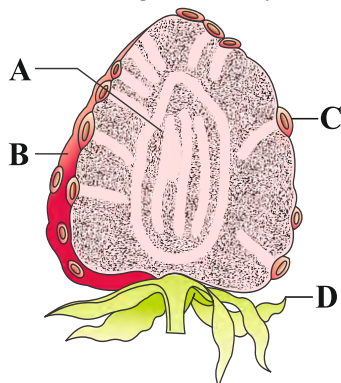
- (1) By producing large amounts of nectar.
- (2) By having brightly coloured petals.
- (3) By secreting foul odours.
- (4) By providing safe places to lay eggs.

- 125.** Which one of the following is not a characteristic of wind-pollinated flowers?

- (1) Light and non-sticky pollen grains
- (2) Well-exposed stamens
- (3) Large, feathery stigma
- (4) Production of nectar

126. The coconut water from a tender coconut represents:
- (1) cellular endosperm.
 - (2) free-nuclear endosperm.
 - (3) the embryo.
 - (4) the pericarp.

127. In the given diagram of a false fruit, identify the label that is involved along with ovary in its formation.



- (1) A
 - (2) B
 - (3) C
 - (4) D
128. The fruits are not fleshy in;
- (1) mango.
 - (2) mustard.
 - (3) guava.
 - (4) orange.
129. Which of the following statements about apomixis are **incorrect**?
- A. It is a form of sexual reproduction.
 - B. It produces seeds without fertilisation.
 - C. Apomictic seeds are disadvantageous for horticulture.
 - D. It is common in some species of *Asteraceae* and grasses.
- Choose the **correct** option.
- (1) A and C
 - (2) B and D
 - (3) B and C
 - (4) A and D
130. The oldest viable seed, excavated from the Arctic Tundra, belongs to the plant:
- (1) *Phoenix dactylifera*.
 - (2) *Lupinus arcticus*.
 - (3) *Ficus*.
 - (4) an orchid.
131. The plant that came into India as a contaminant with imported wheat is;
- (1) cashew.
 - (2) apple.
 - (3) water hyacinth.
 - (4) carrot grass.

132. The generative cell of a mature pollen grain;
- (1) is larger than vegetative cell.
 - (2) lacks the nucleus.
 - (3) is spindle shaped.
 - (4) has abundant food reserves.

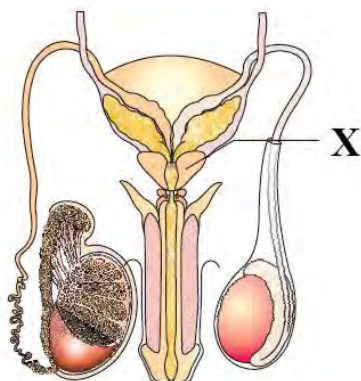
133. Match **List I** with **List II**.

List-I (Structure)		List-II (Develops into)	
(A)	Ovary	(I)	Embryo
(B)	Ovule	(II)	Endosperm
(C)	Zygote	(III)	Fruit
(D)	Primary Endosperm Cell	(IV)	Seed

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-IV, C-I, D-II
 - (2) A-IV, B-III, C-II, D-I
 - (3) A-III, B-IV, C-II, D-I
 - (4) A-IV, B-II, C-III, D-I
134. Consider the following statements.
- A. Pollen grains are rich in nutrients.
 - B. Pollen grains of *Parthenium* are known to cause bronchial afflictions.
 - C. Pollen consumption has been claimed to decrease the performance of athletes and race horses.
 - D. In western countries, a large number of pollen products in the form of tablets and syrups are available in the market.
 - E. The pollen grains represent the female gametophytes.
- Choose the option with all **correct** statements.
- (1) A, B, C, D, E
 - (2) A, B and D only
 - (3) A, B, D and E only
 - (4) C, D and E only
135. Identify the **correct** sequence of embryo development in dicots.
- (1) Proembryo → heart-shaped → globular → mature embryo
 - (2) Proembryo → globular → heart-shaped → mature embryo
 - (3) Proembryo → mature embryo → globular → heart-shaped
 - (4) Heart-shaped → globular → mature embryo → proembryo

136. Select the **correct** option for the labelled part X in the given below diagram.



	Gland	Function
(1)	Seminal vesicle	Release a lubricating, transparent and jelly like substance, which makes approximately 60% part of semen.
(2)	Bulbourethral gland	Release an alkaline fluid that protects sperm by neutralizing urine acids in the urethra.
(3)	Prostate gland	Secretes seminal fluid which makes approximately 60% part of semen.
(4)	Prostate gland	Secretes prostatic fluid that contain several substances.

137. Read the statements (A-E) on spermatogenesis and select the **correct** option:

- Spermatogonia are diploid cells found inside seminiferous tubules.
 - Sertoli cells are involved in nutrient supply to developing sperms.
 - Leydig cells are present inside seminiferous tubules.
 - Spermiation is the process by which mature sperms are released into the lumen of seminiferous tubules.
 - Androgens stimulate spermatogenesis.
- Only A, B, D, E are correct
 - Only B, C, D are correct
 - A, B, C, D are correct
 - Only C, D, E are correct

138. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Zona pellucida	(I)	Diploid stem cells that multiply by mitosis in foetal ovary
(B)	Graafian follicle	(II)	Mature follicle with large fluid-filled antrum
(C)	Oogonia	(III)	Non-cellular glycoprotein layer surrounding the oocyte
(D)	Primary oocyte	(IV)	Oocyte arrested in prophase-I of meiosis

Choose the **correct** answer from the options given below:

- A-II, B-III, C-I, D-IV
- A-III, B-II, C-I, D-IV
- A-III, B-IV, C-I, D-II
- A-III, B-II, C-IV, D-I

139. Which of the following hormones is **not** secreted by placenta?

- hCG
- Estrogen
- hPL
- Prolactin

140. Which of the following statements is **not** correct regarding the distinctive features of human oogenesis compared to spermatogenesis?

- Oogenesis initiates during embryonic development and arrests at metaphase I until puberty.
- Each meiotic division in oogenesis is asymmetric, resulting in polar body formation and retention of cytoplasmic volume in the developing oocyte.
- Oogenesis produces a single mature ovum from each primary oocyte, while spermatogenesis yields four functional sperm from a primary spermatocyte.
- The second meiotic division in oogenesis completes only upon fertilization by sperm.

141. The finger-like projections that appear on the trophoblast after implantation are known as:

- Fimbriae
- Inner cell mass
- Chorionic villi
- Umbilical cord

142. Given below are two statements:

Statement I: The surge in luteinizing hormone (LH) is both necessary and sufficient for triggering ovulation in the human female.

Statement II: Follicle Stimulating Hormone (FSH) exclusively stimulates follicular maturation throughout the cycle, independent of LH.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

143. Consider the following statements regarding the menstrual cycle and choose the **incorrect** statements:

- A. High level of FSH and LH directly inhibit the thickening of endometrium.
- B. Follicular phase includes maturation of the Graafian follicle.
- C. At the end of follicular phase, ovulation takes place.
- D. Secretory phase is regulated by estrogen.
- E. Follicular phase ends with menstruation if fertilization does not occur.

- (1) A, B and C
- (2) A and C only
- (3) D and E only
- (4) A, D and E

144. A 24-year-old male, presents with infertility and reduced sperm count. Laboratory studies reveal normal serum LH and FSH but substantially lowered testosterone. Which part of the testicular architecture is most likely damaged, and what additional cellular effects would be observed?

- (1) Damage to Leydig cells, resulting in impaired androgen production and decreased Sertoli cell support for germ cell maturation.
- (2) Destruction of Sertoli cells, causing direct inhibition of spermatogenesis with elevated testosterone.
- (3) Atrophy of vas deferens, affecting only sperm transport with normal testosterone.
- (4) Destruction of Sertoli cells resulting in impaired androgen production.

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The corpus luteum is essential for the maintenance of pregnancy in its initial weeks in humans.

Reason R: The corpus luteum synthesizes progesterone, which supports vascularization and secretory activity in the endometrium, facilitating implantation and nutrition for the developing embryo.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.

146. The first meiotic division in oogenesis results in:

- (1) Two haploid gametes
- (2) One primary oocyte and one polar body
- (3) One secondary oocyte and one polar body
- (4) Two diploid cells

147. A female patient shows irregular menstruation and difficulty in conceiving. Ultrasound reveals non-maturing follicles, and hormone assay shows low FSH. Most likely, the abnormality is due to dysfunction of:

- (1) Pituitary gland.
- (2) Corpus luteum.
- (3) Thyroid gland.
- (4) Placenta.

148. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Morula	(I)	Outer layer of blastocyst which gets attached to the endometrium
(B)	Blastocyst	(II)	Inner layer of blastocyst that gets differentiated as the embryo
(C)	Trophoblast	(III)	8-16 cell embryo
(D)	Inner cell mass	(IV)	Characterised by formation of a cavity known as blastocoel

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I D-II
- (2) A-IV, B-III, C-I, D-II
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-I, C-II, D-III

149. Choose **correctly** paired match from the given options.

- (1) Rete testis → Long-term storage and physiological maturation of spermatozoa
- (2) Vasa efferentia → Network that connects seminiferous tubules to rete testis
- (3) Epididymis → Short ducts that drain sperm from testis into urethra
- (4) Vas deferens → Transport of sperm from epididymis to ejaculatory duct during ejaculation

150. Read the following statements about the female reproductive system and choose the **correct** option:

- A. Ovaries produce ova and steroid hormones.
 - B. Fallopian tube has infundibulum, ampulla, and isthmus.
 - C. Endometrium undergoes cyclical changes during menstrual cycle.
 - D. Cervix connects uterus to vagina.
 - E. Hymen is a valid indicator of virginity in females.
- (1) Only A, B and C are correct
 - (2) Only B, C, D and E are correct
 - (3) Only C, D and E are correct
 - (4) Only A, B, C and D are correct

151. Which of the following is **not** an accessory reproductive gland in males?

- (1) Seminal vesicles
- (2) Seminiferous tubules
- (3) Bulbourethral glands
- (4) Prostate gland

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

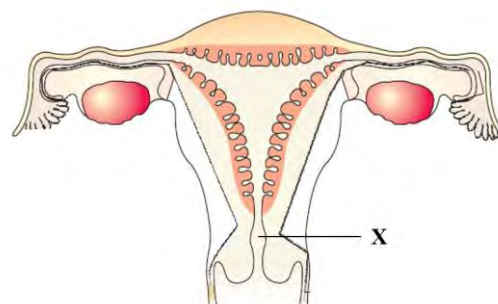
Assertion A: Gonadotropin releasing hormone (GnRH) from the hypothalamus stimulates the anterior pituitary to release FSH and LH.

Reason R: Testosterone, produced by Leydig cells, exerts negative feedback on secretion of LH, FSH and GnRH.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

153. Which of the following **correctly** explain the function of labelled part **X** in given below diagram of female reproductive system?



- (1) Collection of ovum released at ovulation.
- (2) Common site of fertilization of ovum by sperm
- (3) Narrow passage conducting early embryo toward uterus
- (4) Forms part of the birth canal and regulates sperm passage into uterus

154. In lactation, colostrum is important because:

- (1) Contains high amount of fats.
- (2) Rich in antibodies.
- (3) Provides maximum carbohydrates for growth.
- (4) Secreted by pituitary.

155. Consider the following four statements (A-D) and select the **correct** option stating which ones are true (T) and which ones are false (F).

- A. The scrotum maintains the testes at a temperature approximately 2.5°C lower than the core body temperature, optimizing conditions for spermatogenesis.
- B. Spermatogonia are diploid progenitors that initiate germinal cell division
- C. The proliferative phase is marked by rapid mitotic activity in the endometrial stroma, driven by rising progesterone levels.
- D. The cyclical changes in the endometrium are strictly regulated by feedback loops between the hypothalamus, pituitary, and ovaries.

	A	B	C	D
(1)	T	T	F	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	T	T	F	F

156. Which of the following statement about fertilization is **incorrect**?

- (1) Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.
- (2) Only one sperm fertilizes ovum
- (3) Acrosomal enzymes facilitate entry
- (4) Sperm and ovum fuse in uterus

157. Consider the following statements:

Statement I: In a normal menstrual cycle with successful fertilization and implantation, degeneration of the corpus luteum is prevented by human chorionic gonadotropin (hCG).

Statement II: The hCG can be detected in maternal blood and urine and forms the basis of most pregnancy tests.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

158. Match List-I with List-II.

List-I		List-II	
(A)	Spermiogenesis	(I)	Formation of female gamete
(B)	Spermiation	(II)	First menstruation
(C)	Oogenesis	(III)	Maturation of spermatids
(D)	Menarche	(IV)	Release of sperms into lumen of seminiferous tubule

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-III, B-IV, C-I, D-II
- (3) A-III, B-I, C-IV, D-II
- (4) A-III, B-IV, C-II, D-I

159. Which of the following statements are **true** regarding human sperm structure?

- A. The acrosome contains hydrolytic enzymes.
- B. The middle piece has abundant mitochondria.
- C. Head contains diploid chromosomal material.
- D. Tail provides motility and energy for movement.
- E. Plasma membrane surrounds the head region only.

- (1) A, B and E only
- (2) C and D only
- (3) A and B only
- (4) A, B, C, D and E

160. Match List-I with List-II.

List-I (Reproductive Event)		List-II (location/feature)	
(A)	Spermatogenesis	(I)	Ovary
(B)	Oogenesis	(II)	Ampullary region of fallopian tube
(C)	Fertilisation	(III)	Seminiferous tubule
(D)	Implantation	(IV)	Uterus

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-I, C-II, D-IV
- (4) A-III, B-I, C-IV, D-II

161. Given below are two statements:

Statement I: Maintenance of hygiene and sanitation during menstruation is very important.

Statement II: During menstruation it is not necessary to change pads after every 4-5 hours.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

162. Which of the following statements is **incorrect**?

- (1) The hormone oxytocin is a primary stimulator of uterine muscle contractions.
- (2) The foetal ejection reflex triggers release of oxytocin from the maternal pituitary.
- (3) The mammary glands start producing milk towards the end of pregnancy by the process called lactation.
- (4) A surge in Luteinizing Hormone (LH) is required to start the process of labor.

163. Match List-I with List-II.

List-I (Hormone)		List-II (Function)	
(A)	Estrogen	(I)	Regulates menstrual cycle and secondary sexual characters
(B)	Progesterone	(II)	Stimulates growth and maturation of ovarian follicles
(C)	Follicle stimulating hormone	(III)	Causes uterine contractions during childbirth
(D)	Oxytocin	(IV)	Maintains endometrium and supports pregnancy

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-I, D-IV
- (4) A-I, B-IV, C-III, D-II

164. Choose **incorrect** statement regarding the female external genitalia?

- (1) Mons pubis is a cushion of fatty tissue covered by skin and pubic hair.
- (2) Labia majora are paired folds of tissue under the labia minora.
- (3) The opening of the vagina is often covered partially by a membrane called hymen.
- (4) The clitoris is a tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening.

165. Read following statements:

- A. FSH acts on Sertoli cells and stimulates factors for spermiogenesis.
- B. LH acts on Leydig cells, stimulating androgen secretion.
- C. Androgens stimulate spermatogenesis.
- D. Testosterone and follicle stimulating hormone both are androgens.
- E. Spermiogenesis is not a part of spermatogenesis.

How many of the above statements are **not** true?

- (1) Three only (2) Four only
- (3) Two only (4) All five

166. Match List-I with List-II.

List-I		List-II	
(A)	Fimbriae	(I)	Present inside the seminiferous tubule
(B)	Infundibulum	(II)	Help collect the ovum after ovulation
(C)	Sertoli cells	(III)	Secrete testosterone
(D)	Interstitial cells	(IV)	Funnel-shaped opening of oviduct near the ovary

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-III, B-IV, C-II, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

167. Read the following statements regarding uterus.

- A. The uterus is single and it is also called womb.
- B. The shape of the uterus is like an inverted pear.
- C. It is supported by tendons attached to the pelvic wall.
- D. The uterus opens into vagina through a narrow cervix.

Which of the above statement is **not** correct?

- (1) A and B (2) C and D
- (3) C only (4) D only

168. Following fertilization, the newly formed zygote undergoes a series of developmental changes as it travels to the uterus. What is the **correct** chronological sequence of these early embryonic stages?

- (1) Zygote → Blastocyst → Morula → Embryo
- (2) Zygote → Morula → Blastocyst → Embryo
- (3) Morula → Zygote → Blastocyst → Embryo
- (4) Blastocyst → Embryo → Zygote → Morula

169. Which of the following option **correctly** represent the average duration of a healthy human pregnancy?

- (1) 46 - 50 weeks (2) 36 - 40 weeks
- (3) 10- 12 months (4) 30-34 weeks

170. Regarding fertilization and implantation, read these statements:

- A. Fertilization is the process of male and female gamete fusion to form a diploid zygote.
- B. Zona pellucida blocks entry of multiple sperms.
- C. Morula is formed after blastocyst stage.
- D. Implantation occurs in the outermost uterine wall.
- E. Sex determination occurs at the time of gamete formation.

Which of the above statements is/are **not** correct?

- (1) A and B only (2) C and D only
- (3) A, B and D only (4) C, D and E only

171. Given below are two statements:

Statement I: The penis is the male external genitalia which is made up of special tissue that helps in erection of the penis to facilitate gametogenesis.

Statement II: Seminal plasma is rich in fructose, calcium and certain enzymes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

172. Read the following statements about placenta:

- A. It allows exchange of gases and wastes between embryo and maternal blood.
- B. It is connected to foetus via umbilical cord.
- C. It acts as an endocrine tissue and produces several hormones such as oxytocin.
- D. Placenta is derived only from maternal tissue.
- E. It allows the mixing of maternal and foetal blood to ensure maximum nutrient exchange.

Which of the above statements is/are **correct**?

- (1) A, B and C only (2) B and D only
- (3) D and E only (4) A and B only

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: A high sperm count, such as 300 million sperms per ejaculate, is the sole factor determining male fertility.

Reason R: Even with a high sperm count, fertility can be compromised if a large percentage of sperm have structural abnormalities or are not actively motile.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

174. Which of the following options most **accurately** describes a primary and defining characteristic of the endometrium, the inner lining of the uterus?
- (1) The ability to perform powerful muscle contractions during labor.
 - (2) Undergoing hormonally-driven, cyclical changes throughout the menstrual cycle.
 - (3) The production and release of female gametes (oocytes).
 - (4) The secretion of fluid for lubrication during coitus.

175. Which of the following structures contribute secretions that are essential for the maturation and motility of sperm, and form the seminal plasma?

I. Epididymis II. Vas deferens
III. Seminal vesicle IV. Prostate
V. Seminiferous tubules

Select the **correct** option from the choices below:

- (1) I and II only
- (2) II, III and IV only
- (3) I, II, III and IV only
- (4) V only

176. Choose **incorrect** statement from the given options.

- (1) The process of formation of a mature female gamete is called oogenesis.
- (2) A functional mammary gland is characteristic of all female mammals.
- (3) Perimetrium is the outermost layer of uterus which comprises of smooth muscle.
- (4) The last part of the oviduct, isthmus has a narrow lumen and it joins the uterus.

177. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Corpus luteum	(I)	Multiple layers of granulosa cells; formation of theca begins
(B)	Primary follicle	(II)	Presence of fluid-filled antrum; well-developed theca layers
(C)	Secondary follicle	(III)	Single layer of granulosa cells around primary oocyte
(D)	Tertiary follicle	(IV)	Secretes progesterone

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-IV, B-II, C-I, D-III

178. Which of the following represents the **correct** pathway for the transport of milk in the mammary gland?

- (1) Alveoli → Mammary duct → Mammary tubules → Mammary ampulla → Lactiferous duct
- (2) Alveoli → Mammary tubules → Mammary duct → Mammary ampulla → Lactiferous duct
- (3) Mammary tubules → Alveoli → Mammary ampulla → Mammary duct → Lactiferous duct
- (4) Alveoli → Mammary ampulla → Mammary duct → Lactiferous duct → Nipple

179. Regarding embryonic development, which of the following statements is/are **correct**?

- A. Embryo's heart is formed by the end of month one.
- B. By the end of 4 weeks, major organ systems are formed.
- C. Limb buds and digits appear by month two.
- D. Appearance of hair on head is seen by third month.
- E. Eyelids and eyelashes are formed by 24 weeks.

- (1) A, C and E only
- (2) A, C, D and E only
- (3) B and D only
- (4) C, D and E only

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: There is a remarkable difference between the reproductive events in human males and females regarding the duration of gamete formation.

Reason R: Sperm formation continues in males even in old age, but the formation of ovum ceases in women around the age of sixty years.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true, and R is the correct explanation of A.
- (2) Both A and R are true, but R is not the correct explanation of A.
- (3) A is true, but R is false.
- (4) A is false, but R is true.



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____